Review article: Numerical Methods for Engineering Quantum Computers

Liu, Shu
Mathematics, Florida State University

Georgiadou, Antigoni
Oak Ridge National Laboratory

Dong, Spancer
Oak Ridge National Laboratory

Kubischta, Eric
Mathematics, Florida State University

Melton, Daniel
Mathematics, Florida State University

Guenther, Stefanie
Lawrence Livermore National Laboratory

Jin, Shi
Mathematics, Shanghai Jiao Tong University

Lin, Lin
Mathematics, U.C. Berkeley

Lotshaw, Philip
Oak Ridge National Laboratory

Petersson, Anders
Lawrence Livermore National Laboratory

Sussman, Mark
Mathematics, Florida State University

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ABSTRACT: In the article by Herrmann et al, "Quantum utility – definition and assessment of a practical quantum advantage," the "Application Readiness Level" (ARL) is defined: (ARL1) "concept with unknown potential," (ARL2) "beneficial in small idealized systems," (ARL3) "utility indicated by

theory or resource estimations," (ARL4) "simulated utility demonstration," and (ARL5) "utility demonstration on quantum hardware." At the present, practitioners have reached ARL3 with the Variational Quantum Eigenvalue (VQE) algorithm. This seems to indicate that there is still much work to be done in order to reach ARL4 or ARL5. In this review article, the latest activities from the computational "digital twin" perspective are reported on in which scientists are working hard to bring practitioners out of the Noisy Intermediate Scale Quantum (NISQ) era into an era of "Quantum supremacy." In otherwords this review article discusses digital twin activities that help to advance from ARL3 to ARL4, or even ARL5. The three main categories in this review article are (1) Quantum algorithms for benchmarking quantum computers (to include emulator algorithms like Qiskit), (2) Optimization, Control, inverse problems, and exploratory simulation algorithms for Schrodinger's equation, and (3) Quantum Error correction algorithms and surface codes.

1 Introduction

The motivation for this review article is the fact that Moore's law is no longer applicable[83]. As quoted from Shalf[102], "Moore's Law [1] is a techno-economic model that has enabled the IT industry to double the performance and functionality of digital electronics roughly every 2 years within a fixed cost, power and area. This expectation has led to a relatively stable ecosystem (e.g. electronic design automation tools, compilers, simulators and emulators) built around general-purpose processor technologies, such as the ×86, ARM and Power instruction set architectures. However, within a decade, the technological underpinnings for the process that Gordon Moore described will come to an end, as lithography gets down to atomic scale. At that point, it will be feasible to create lithographically produced devices with dimensions nearing atomic scale, where a dozen or fewer silicon atoms are present across critical device features, and will therefore represent a practical limit for implementing logic gates for digital computing [2]. Indeed, the ITRS (International Technology Roadmap for Semiconductors), which has tracked the historical improvements over the past 30 years, has projected no improvements beyond 2021, as shown in figure 1, and subsequently disbanded, having no further purpose. The classical technological driver that has underpinned Moore's Law for the past 50 years is failing [3] and is anticipated to flatten by 2025, as shown in figure 2. Evolving technology in the absence of Moore's Law will require an investment now in computer architecture and the basic sciences (including materials science), to study candidate replacement materials and alternative device physics to foster continued technology scaling."

References one through three in the above quote correspond to the following references respectively: [83][80][81].

As further motivation for this review article, in Figure 3 of the article by Shalf[102], a roadmap is provided for possible paths forward when the density of circuits on classical computers exceeds a critical value. There are three categories: (a) roadmap for the next ten years, (b) 20 years, and (c) "Decades beyond exascale," "New Models of Computation." For category (c), Shalf[102] refers to the following prospective technology: (i) approximate computing, (ii) adiabatic reversible, (iii) Analog, (iv) Neuromorhic, and (v) quantum[28, 29]. Of the "new models of computation" listed in Shalf's article[102], this review article will address the "Numerical Methods for Engineering Quantum Computers" aspects associated with the emerging "quantum computing" paradigm. As outlined by Gamble[30], the development of reliable quantum computers will have a tranformative effect on computer technology. That being said, right now we are in the "Noisy Intermediate-Scale Quantum"[15] (NISQ) era. The purpose of this review article is to bring to the forefront the current research being done, from the computational perspective, in order to move beyond the NISQ era.

2 Quantum algorithms and Quantum emulators for benchmarking quantum computers

Quantum Emulators:

1. (IBM) "Quantum Computing with Qiskit" (2024)[51]
2. "Intel Quantum Simulator: a cloud-ready high-performance simulator of quantum circuits" (2020)[35]
3. (Microsoft) "LIQUid: A Software design architecture and domain specific language for quantum computing" (2014)[112]
4. "QuESTlink—Mathematica embiggened by a hardware-optimised quantum emulator" (2020)[55]
5. (Matlab) "QCLAB: A Matlab Toolbox for Quantum Computing" (2025)[58]
6. "PennyLane-Lightning MPI: A massively scalable quantum circuit simulator based on distributed computing in CPU clusters" (2025)[57]
7. (Google) "qsim"(2020)[109]
8. (nVidea) "cuQuantum SDK: A High-Performance Library for Accelerating Quantum Science" (2023)[8]
9. "Numerical recipes in quantum information theory and quantum computing: an adventure in FORTRAN 90"(2021)[96]
10. Nguyen et al(2022)[87] "Tensor network quantum virtual machine for simulating quantum circuits at exascale"
11. Zhang et al (2024)[119], "Quantum circuit simulation with fast tensor decision diagram"

12. Viamontes et al (2004)[111], “High-performance QuIDD-based simulation of quantum circuits.”

13. Ito and Yoshioka (2024)[50], “Quantum computer simulation on HPC.”

14. Xu et al (2024)[116], “Atlas: Hierarchical partitioning for quantum circuit simulation on gpus”

15. Zhang et al (2021)[117], “HyQuas: hybrid partitioner based quantum circuit simulation system on GPU”

16. Li et al (2021)[70], “Sv-sim: scalable pgas-based state vector simulation of quantum circuits”

17. Kolarovszki et al (2025)[67], “Piquasso: A photonic quantum computer simulation software platform”

Quantum Algorithms for benchmarking Quantum Computers:

1. Kitaev (1995)[62], “Quantum measurements and the Abelian stabilizer problem” (quantum phase estimation)

2. Shor (1999)[105], “Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer”

3. Callison and Chancellor (2022)[15] “Hybrid quantum-classical algorithms in the noisy intermediate-scale quantum era and beyond”

4. Jin et al (2024)[52], “Quantum Simulation for Quantum Dynamics with Artificial Boundary Conditions.”

5. S. Jin, N. Liu, and Y. Yu (2024)[54], “Quantum Circuits for the heat equation with physical boundary conditions via Schrodingerisation”

6. Cerezo et al (2021)[16] “Variational quantum algorithms”

7. Lotshaw et al (2021)[77] “Simulations of frustrated Ising Hamiltonians using quantum approximate optimization”

8. Nielsen and Chuang (2010)[88] “Quantum computation and quantum information” (quantum Fourier Transform and quantum phase estimation)

9. Ding and Lin (2023)[22] “Simultaneous estimation of multiple eigenvalues with short-depth quantum circuit on early fault-tolerant quantum computers”

10. Biamonte et al (2017)[10] “Quantum machine learning”

11. Benedetti et al (2019)[9] “Parameterized quantum circuits as machine learning models”

12. Khait et al(2023)[59] ”Variational quantum eigensolvers in the era of distributed quantum computers”

13. Sawaya et al (2023)[99], “HamLib: A Library of Hamiltonians for Benchmarking Quantum Algorithms and Hardware”
14. Georgadou et al(2024)[33] Hele-Shaw simulation on quantum computer.
15. Kim et al (2023)[61], “Evidence for the utility of quantum computing before fault tolerance”
16. Zoufal et al (2019)[122], “Quantum generative adversarial networks for learning and loading random distributions”
17. Nghiem and Tzu-Chieh (2023)[86], “Quantum algorithm for estimating largest eigenvalues”
18. Cadi et al (2024)[14], “Folded spectrum vqe: A quantum computing method for the calculation of molecular excited states”
19. Tilly et al (2022)[110], “The variational quantum eigensolver: a review of methods and best practices”
20. Qing and Xie (2023)[94], “Use VQE to calculate the ground energy of hydrogen molecules on IBM Quantum”
21. Ryu et al (2021)[98], “A Variational Quantum Algorithm for Ordered SVD”
22. Shirakawa et al (2024)[103], “Automatic quantum circuit encoding of a given arbitrary quantum state”
23. Harrow et al (2009)[41], “Quantum algorithm for linear systems of equations”
24. Daribayev et al (2023)[19], “Implementation of the hhl algorithm for solving the poisson equation on quantum simulators.”
25. Bravo-Prieto et al (2023)[12], ”Variational quantum linear solver”
26. Gilyen et al (2019)[31], “Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics”
27. Gundlach et al (2025)[36], “Quantum Advantage in Computational Chemistry?”
28. Ko et al (2023)[65], “Implementation of the density-functional theory on quantum computers with linear scaling with respect to the number of atoms”
29. Jin et al (2025)[53], “Quantum preconditioning method for linear systems problems via Schrodingerization”
30. Itani et al (2024)[49], “Quantum algorithm for lattice Boltzmann (QALB) simulation of incompressible fluids with a nonlinear collision term”

31. Hibat-Allah et al (2024)[46], "A framework for demonstrating practical quantum advantage: comparing quantum against classical generative models"
32. Herrmann et al (2023)[45], "Quantum utility–definition and assessment of a practical quantum advantage"
33. Lotshaw et al (2023)[75], "Modeling noise in global Molmer-Sorensen interactions applied to quantum approximate optimization."
34. Di Bartolomeo et al (2023)[21], "Noisy gates for simulating quantum computers."
35. De Luo et al (2025)[78], "Quantum simulation of bubble nucleation across a quantum phase transition."
36. Maciejewski et al (2024)[79], "Improving Quantum Approximate Optimization by Noise-Directed Adaptive Remapping."
37. Seki et al (2025)[100], "Simulating Floquet scrambling circuits on trapped-ion quantum computers"
38. Rakyta et al (2024)[95], "Highly optimized quantum circuits synthesized via data-flow engines"
39. Popovici et al (2025)[92], "A Closeness Centrality-based Circuit Partitioner for Quantum Simulations"
40. Hsu et al (2024)[47], "Toward cost-effective quantum circuit simulation with performance tuning techniques"
41. Park et al (2022)[91], "SnuQS: scaling quantum circuit simulation using storage devices"

3 Optimization, Control, inverse problems, and exploratory simulation algorithms for Schrodinger's equation

1. Feit et al (1982)[27], "Solution of the Schrödinger equation by a spectral method."
2. Bao et al (2002)[4], "On Time-Splitting Spectral Approximations for the Schrödinger Equation in the Semiclassical Regime."
3. Lin et al (2016)[74], "Approximating spectral densities of large matrices."
4. Zhang et al (2017)[118], "Observation of a many-body dynamical phase transition with a 53-qubit quantum simulator."

5. Han et al (2019)[39], “Solving many-electron Schrödinger equation using deep neural networks.”

6. Günther et al (2021) [38], “Quandary: An open-source C++ package for high-performance optimal control of open quantum systems.” the article by Lidar (2019)[72] (“Lecture notes on the theory of open quantum systems”) gives the derivation of the model for open quantum systems.

7. Günther and Petersson (2023) [37], “A practical approach to determine minimal quantum gate durations using amplitude-bounded quantum controls.”

8. Fei et al (2025)[25], “Switching time optimization for binary quantum optimal control.”

9. Fei et al (2023)[26], “Binary control pulse optimization for quantum systems.”

10. Kanai et al (2024)[56], “Single-Electron Qubits Based on Quantum Ring States on Solid Neon Surface.”

11. Hermann et al (2020)[44], “Deep-neural-network solution of the electronic Schrödinger equation.”

12. Andrade et al (2022)[2], “Engineering an effective three-spin Hamiltonian in trapped-ion systems for applications in quantum simulation.”

13. Wille et al (2018)[113], “Computer-aided design for quantum computation.”

14. He et al (2024)[43], “Efficient Optimal Control of Open Quantum Systems”

15. Nusseler et al (2020) [90], “Efficient simulation of open quantum systems coupled to a fermionic bath.”

16. Selsto and Kvaal (2010)[101], “Absorbing boundary conditions for dynamical many-body quantum systems.”

17. Sawaya et al (2023)[99], “HamLib: A Library of Hamiltonians for Benchmarking Quantum Algorithms and Hardware.”

18. Grivopoulos (2005)[34], “Optimal control of quantum systems.”

19. Hangleiter et al (2024)[40], “Robustly learning the Hamiltonian dynamics of a superconducting quantum processor.”

20. Shukrinov et al (2016)[106], “Modeling of LC-shunted intrinsic Josephson junctions in high-Tc superconductors.”

21. Sun and Galperin (2025)[108], “Control of open quantum systems: The nonequilibrium Green’s function perspective.”

22. Zhang et al (2004)[120], “Numerical solutions of the Schrodinger equation for the ground lithium by the finite element method.”
23. He and Wu (2024)[42], “Zeroing neural dynamics solving time-variant complex conjugate matrix equation”
24. Liao et al (2020)[71], “Modified gradient neural networks for solving the time-varying Sylvester equation with adaptive coefficients and elimination of matrix inversion”
25. Lin Lin (2025)[73], “Dissipative preparation of many-body quantum states: Toward practical quantum advantage”
26. Kodali and Motamarri (2025)[66], “Finite-element methods for noncollinear magnetism and spin-orbit coupling in real-space pseudopotential density functional theory”
27. Barrera et al (2025)[6], “The Moving Born-Oppenheimer Approximation”
28. Nieminen et al (2023)[89], “Atomistic modeling of a superconductor–transition metal dichalcogenide–superconductor Josephson junction”
29. Mucciolo et al (2025)[84], “Green’s Function Methods for Computing Supercurrents in Josephson Junctions”
30. Williams-Young et al (2023)[114], “Distributed memory, GPU accelerated Fock construction for hybrid, Gaussian basis density functional theory”

4 Quantum Error Correction algorithms and surface codes

1. Shor (1995)[104], “Scheme for reducing decoherence in quantum computer memory.”
2. Anthony-Petersen et al (2024)[3], “A stress-induced source of phonon bursts and quasiparticle poisoning.”
3. Lotshaw et al (2024)[76], “Exactly solvable model of light-scattering errors in quantum simulations with metastable trapped-ion qubits.”
4. Kubischta, Eric and Teixeira, Ian (2023)[69], “Family of quantum codes with exotic transversal gates.”
5. Barends et al (2014)[5], “Superconducting quantum circuits at the surface code threshold for fault tolerance.”
6. Baum et al (2021)[7], “Experimental deep reinforcement learning for error-robust gate-set design on a superconducting quantum computer.”

7. Cheng and Guo (2025)[17], "Modeling correlated-noise in silicon spin qubit device."
8. Putterman et al (2025)[93], "Hardware-efficient quantum error correction via concatenated bosonic qubits"
9. Knill and Laflamme (1997)[64], "Theory of quantum error-correcting codes"
10. Eastin and Knill (2009)[23], "Restrictions on transversal encoded quantum gate sets"
11. Ide et al (2025)[48], "Fault-tolerant logical measurements via homological measurement"
12. Bravyi and Kitaev (2002)[13], "Fermionic quantum computation"
13. Dennis et al (2002)[20], "Topological quantum memory"
14. Kitaev (2003)[63], "Fault-tolerant quantum computation by anyons"
15. Roffe (2019)[97], "Quantum error correction: an introductory guide"
16. Bluvstein et al (2024)[11], "Logical quantum processor based on reconfigurable atom arrays"
17. Marra (2022)[82], "Majorana nanowires for topological quantum computation"
18. Evered et al (2023)[24], "High-fidelity parallel entangling gates on a neutral-atom quantum computer"
19. Wintersperger et al (2023)[115], "Neutral atom quantum computing hardware: performance and end-user perspective"
20. Gonthier et al (2022)[32], "Measurements as a roadblock to near-term practical quantum advantage in chemistry: Resource analysis"
21. Dalton et al (2024)[18], "Quantifying the effect of gate errors on variational quantum eigensolvers for quantum chemistry"
22. Nautrup et al (2019)[85], "Optimizing quantum error correction codes with reinforcement learning"
23. Krinner et al (2022)[68], "Realizing repeated quantum error correction in a distance-three surface code"
24. Stetina and Wiebe (2025)[107], "First-Quantized Quantum Simulation of Non-Relativistic QED with Emergent Topologically Protected Coulomb Interactions"
25. Zou and Hoi-Kwong (2025)[121], "Algebraic Topology Principles behind Topological Quantum Error Correction"

26. Google Quantum (2025)[1], "Quantum error correction below the surface code threshold"

27. Khan et al (2025)[60], "Separate and efficient characterization of state-preparation and measurement errors using single-qubit operations"

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